

NLP Week - 1

L1

Understanding NLP

- lang. → used to express & communicate. Spoken by humans. It is diff. from PL. NL includes → English, Tamil, Hindi, etc.
- charac. & challenges
 - structured words & sounds
 - human has capacity to use complex sys. to express thoughts
 - it helps in exchanging knowledge
 - varies greatly
- NLP → understand, interpret & generate human language
 - (goal) creatⁿ of lang. models (mimic human mind)
 - combines computational linguistics & ML/DL to analyze large amts. of natural lang. data
- applicⁿ → chatbot, sentiment analysis, MT, speech recog.
- NLP → NLU + NLG → generaⁿ (by machine)
 - ↳ understanding (for machine)
- conceptualizaⁿ → formulaⁿ → articulaⁿ
 - complex sys. of NLU & NLG
- linguistic components → phonology, morphology, phonotactics, syntax, semantics, pragmatics
- lang. → computer → lang.
 - ← NLU ← NLG
- challenges → context, ltd reasoning, bias

- human & machine in NLP has several key differences

L2 History of NLP

- NLP → human-like intelligence & interaction
- Translational Memorandum (machine translation)
- rule-based MT (dictionary lookup, word reordering)
 - ↳ ELIZA
- Turing-test → benchmark for machine intelligence
- case grammars → semantic networks → conceptual dependency network
- expert systems (symbolic approaches)
 - ↳ hard-coded rules + ontologies
- statistical models → learn from the data
 - ↳ (ML)
- RNN → Neural networks (language modeling)
 - ↳ predicⁿ of next word based on prev. words
 - ↳ introducⁿ of word embeddings
 - ⇓
 - grouping of similar kind of words
- LSTM & CNNs → Enc. - dec. (seq2seq model)
- finally, attenⁿ & Transformers + increased text corpora.
- Pre-trained LM → fine-tuning

L3

Why it is hard to process the NL?

- NLP has many core technologies & numerous applications
- why NLP is hard? → ambiguity is present at many levels, lot of linguistic diversity (Sapir-Whorf hypothesis - lang. we speak both affects & reflects our view of the world), we have several language families
- low resource lang. → lang. lacking corpora & crafted linguistic resources sufficient for building statistical NLP applications.
- some NLP technologies are resource-rich like NER, Parsing, etc.
- around 6K world lang. are resource-poor languages.
- ambiguity in NL → multiple interpretations for same phrase, it makes the algo. to understand the context.
 1. lexical → words with multiple meanings.
homonyms, polysemy
 2. syntactic → when a sentence can be parsed in more than 1 way due to its structure
↳ attachment, ambiguity
 3. semantic → phrases have multiple meanings owing to quantifier scope & anaphoric ambiguity
 4. pragmatic → diff. sentence interpretations based on context → deictic, speech act, irony (sarcasm)

- challenges in multilingual NLP → diff. in everything for every language. India has 22 official lang. difficult to have unified models.
- challenges with low-resource languages → lack of large annotated datasets & linguistic tools
- lang. variability across speakers → variations in dialects, sociolects, idiolects

L4 Levels of Lang. Processing & Linguistic Fundamentals

- req. multiple levels of processing → phonology, morphology, lexical, syntactic, semantic, discourse, pragmatic
- processing → charac. → word/token → sentence → paragraph / passage → doc. → multiple docs
- phonological → processing of sounds with its patterns
- morphological → structure & "forma" of words with their int. components
→ lemmatization, stemming
- lexical → vocab of lang., meaning & usage on words. (tokenization, lexical semantics, synonym detecⁿ)
- syntactic → arrangement of words to form grammatical sentences
(MT, parser)
- semantic → meaning is constructed & interpreted (sentiment, IR, MT)
NER

- _/_/_
- discourse - how sentences relate to each other in coherence & cohesion
(text summary, MT, dialogue)
coherence → logical flow of ideas
cohesion → grammatical & lexical linking
 - pragmatic → interpretation of meaning in communication (meanings, sarcasm, dynamics)
 - these all levels are impt. and how they interact with each other.